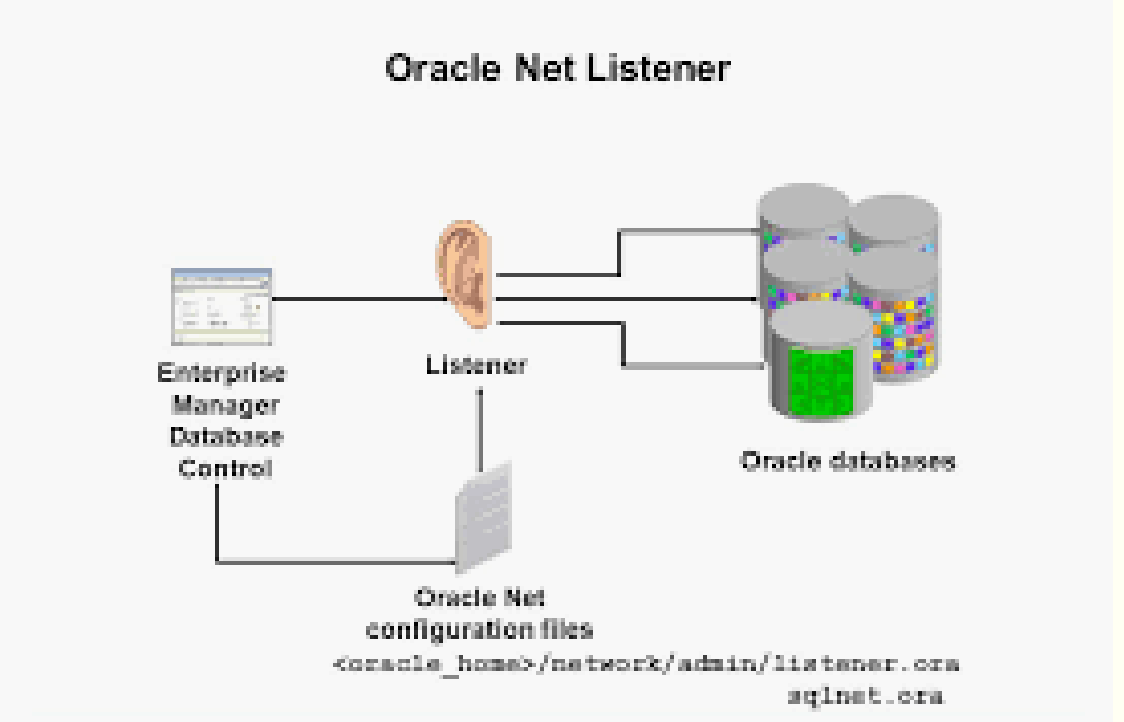
Database Administration & Storage systems – IT4011
Database Administration – IT4040

CONFIGURING DATABASE NETWORK ENVIRONMENT

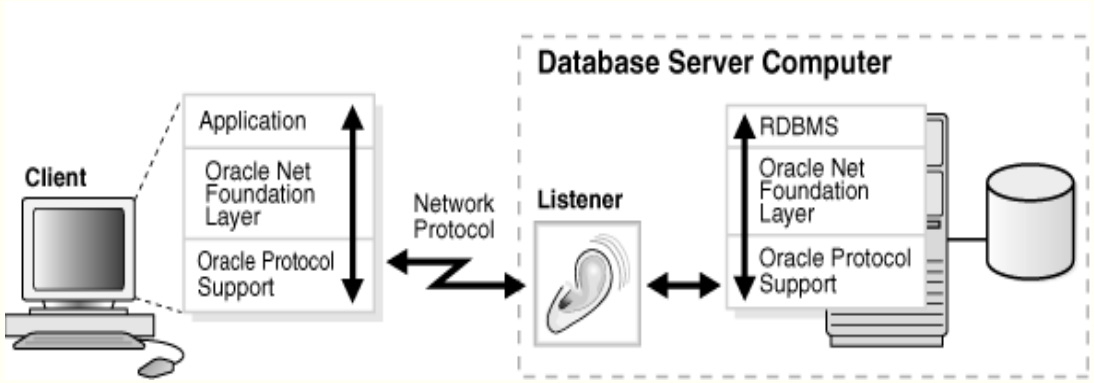


Oracle Net Services Architecture

- **Oracle Net Services** is a suite of networking components that provides enterprise-wide connectivity solutions in distributed, heterogeneous computing environments.
- Oracle Net Services enables a network session from an application to a database instance and a database instance to another database instance.
- Oracle Net Services provides location transparency, centralized configuration and management, and quick installation and configuration

Oracle Net Listener

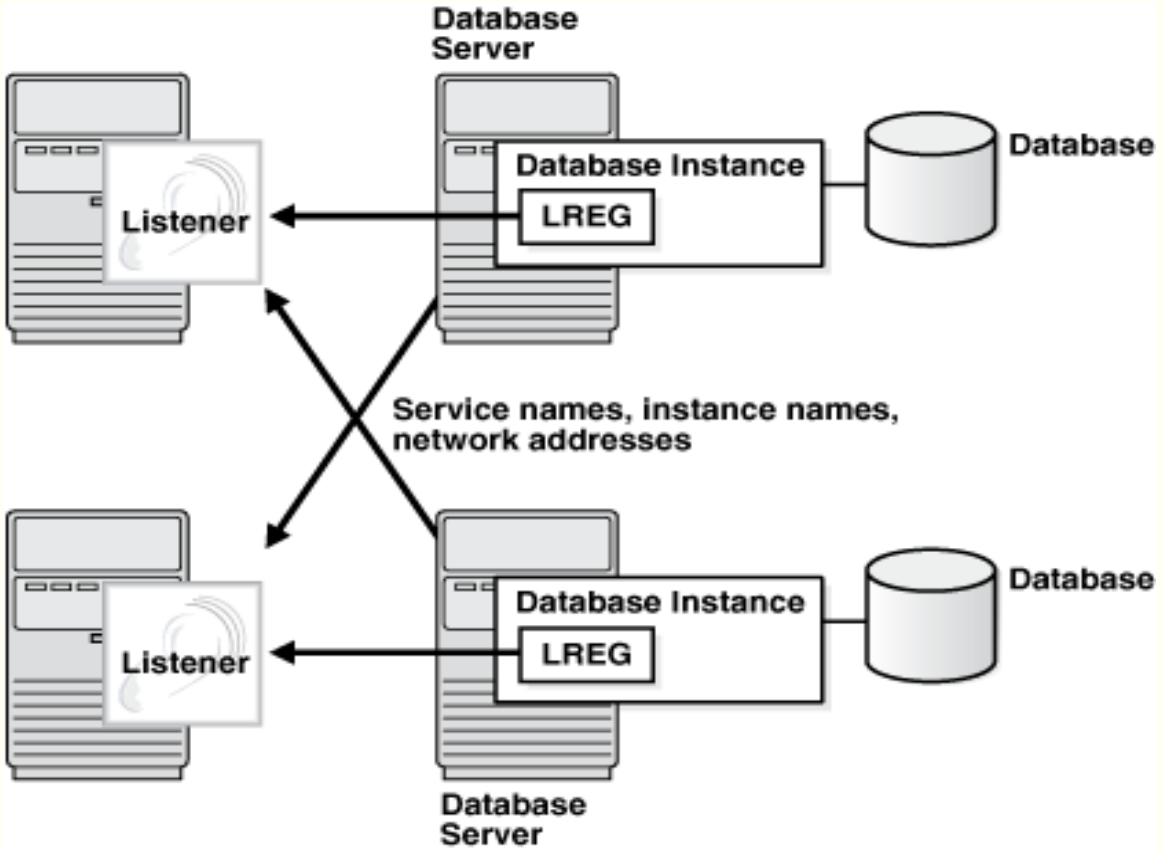
- Listener is a server-side process that listens for incoming client connection requests and manages traffic to the database.
- The instance contacts a listener and establishes a communication pathway to this instance.
- The **database services** running are registered with the listener by LREG process



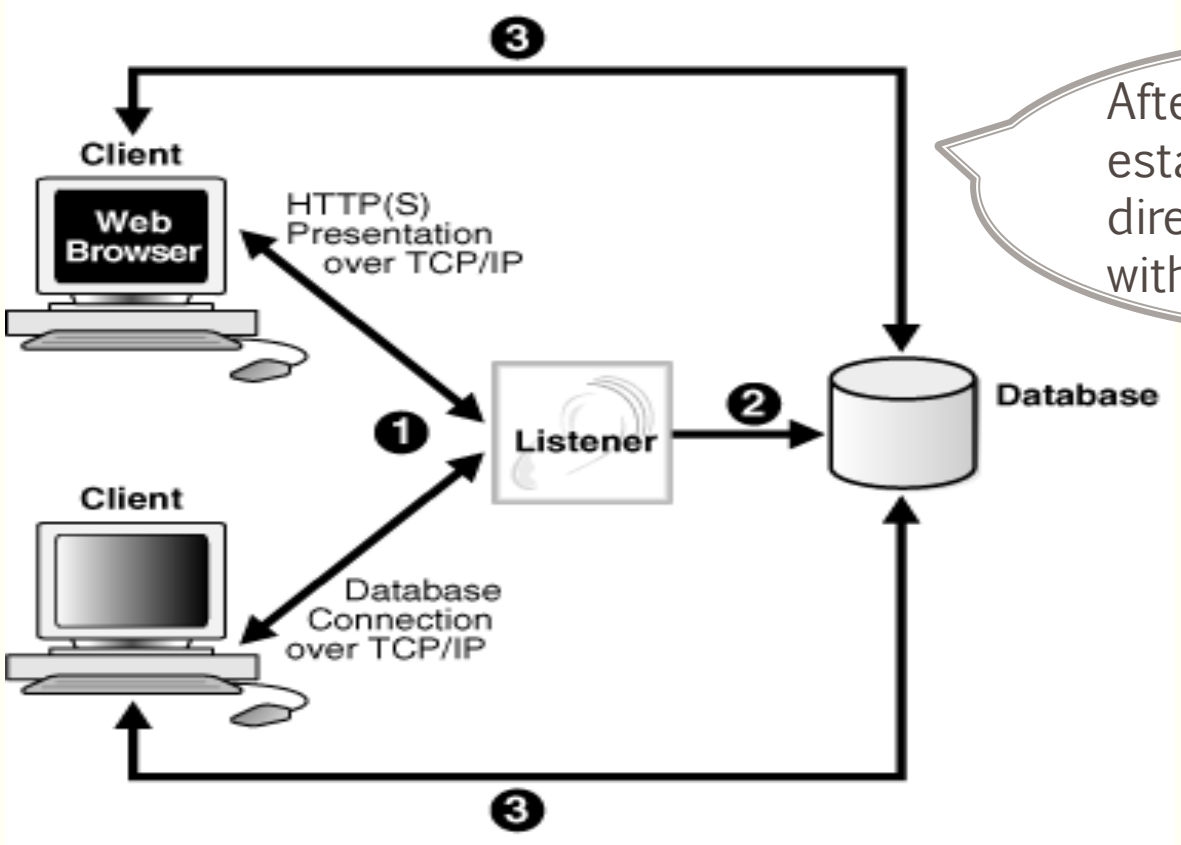
A **service name** is a logical representation of a database, which is the way a database is presented to clients. A single database can be presented as multiple services.

Databases and Listeners

One listener can listen for more than one databases and one database can be listened by a more than one listeners



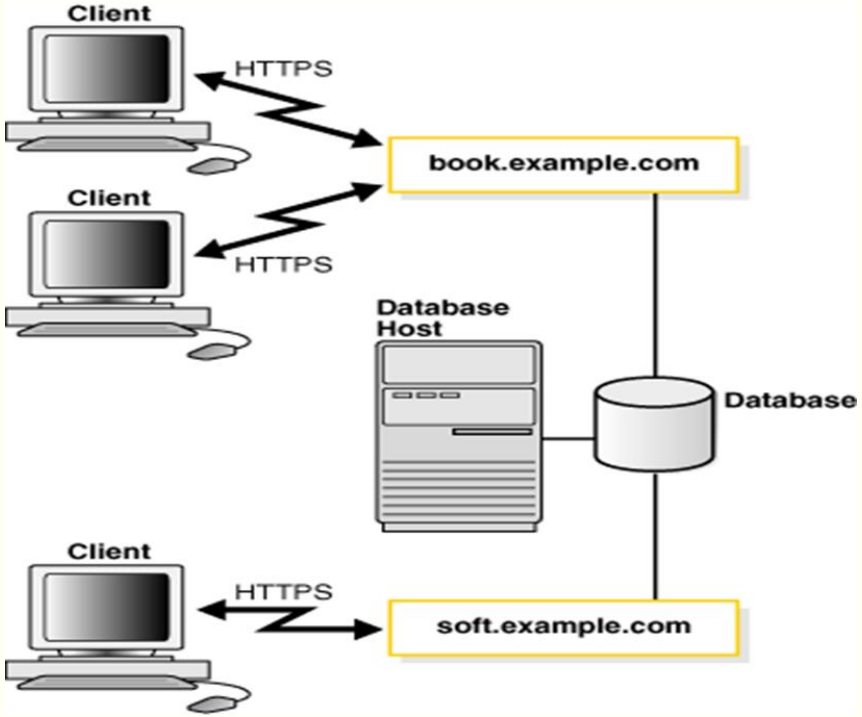
Establishing a Database Connection



After connection is established, client directly communicates with server

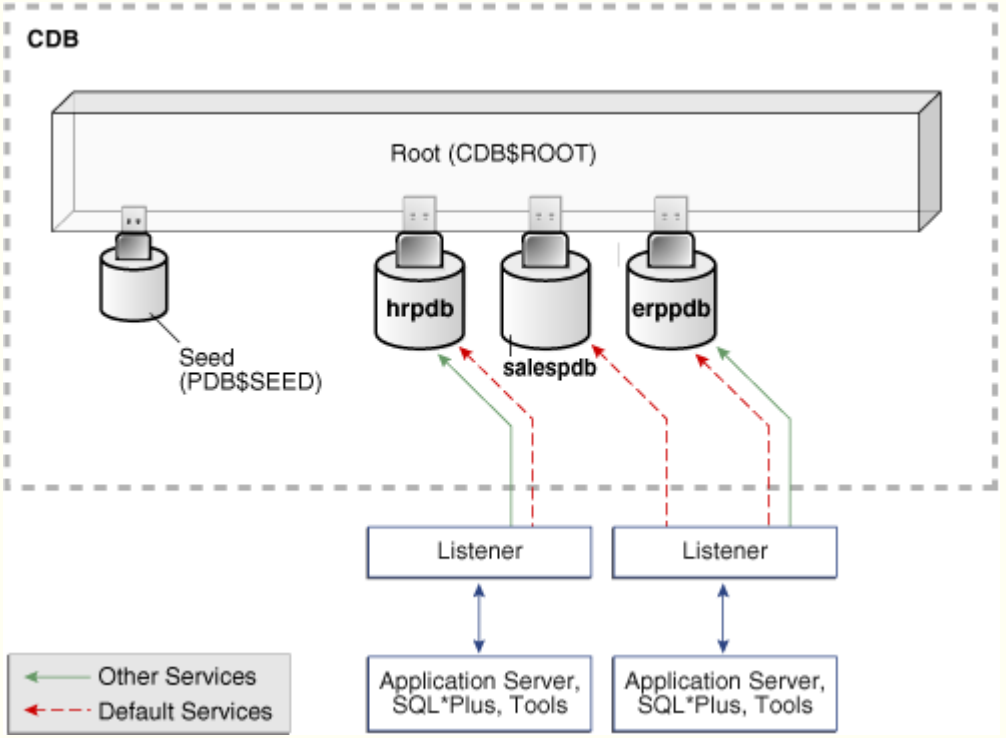
Service Name

- A service name is a logical representation of a service used for client connections. When a client connects to a listener, it requests a connection to a service.
- When a database instance starts, it registers itself with a listener as providing one or more services by name. Thus, the listener acts as a mediator between the client and instances and routes the connection request to the right place.
- Multiple services can be available in a single database



Overview of Services in a CDB

Clients must connect to PDBs using services. A connection using a service name starts a new session in a PDB.



Tools for Configuring & Managing the Oracle Network

- Oracle Net Manager
- Oracle Net Configuration Assistant launched by Oracle Universal Installer
- Command line

Oracle Net Listener Configuration

The default listener configuration file is listener.ora and it is in $ORACLE_HOME/network/admin

An Oracle database get registers with default listener just after starting it

listener.ora file contains following parameters

- Name of the listener
- Protocol addresses that the listener is accepting connection requests on
- Valid nodes that the listener allows to register with the database
- Database services
- Control parameters

Sample Listener.ora File

```
LISTENER=
   (DESCRIPTION=
      (ADDRESS_LIST=
         (ADDRESS=(PROTOCOL=tcp)(HOST=sale-server)(PORT=1521))
         (ADDRESS=(PROTOCOL=ipc)(KEY=extproc))))
SID_LIST_LISTENER =
 (SID_LIST =
   (SID_DESC =
    (GLOBAL_DBNAME = db01)
    (ORACLE_HOME = /u01/app/oracle/product/11.1.0/db_1)
    (SID_NAME = db01)
   )
   (SID_DESC =
    (GLOBAL_DBNAME = dev12c)
    (ORACLE_HOME = /u01/app/oracle/product/12.1.0/dbhome_1)
    (SID_NAME = dev12c)
   )
 )
```

Listener Control Utility

Oracle Net listeners can be controlled with the command-line `lsnrctl` utility (or from EM).

```
$lsnrctl
LSNRCTL for Linux: Version 10.2.0.0.0 on 12-MAY-2005 13:27:51
Copyright (c) 1991, 2004, Oracle. All rights reserved.
Welcome to LSNRCTL, type "help" for information.
LSNRCTL> help
The following operations are available
An asterisk (*) denotes a modifier or extended command:

start               stop                status
services            version             reload
save_config         trace               spawn
change_password     quit                exit
set*                show*
```

Listener Control Utility Syntax

Commands from the listener control utility can be issued from the command line or from the `LSNRCTL` prompt.

UNIX or Linux command-line syntax:

```
$ lsnrctl <command name>
$ lsnrctl start
$ lsnrctl status
```

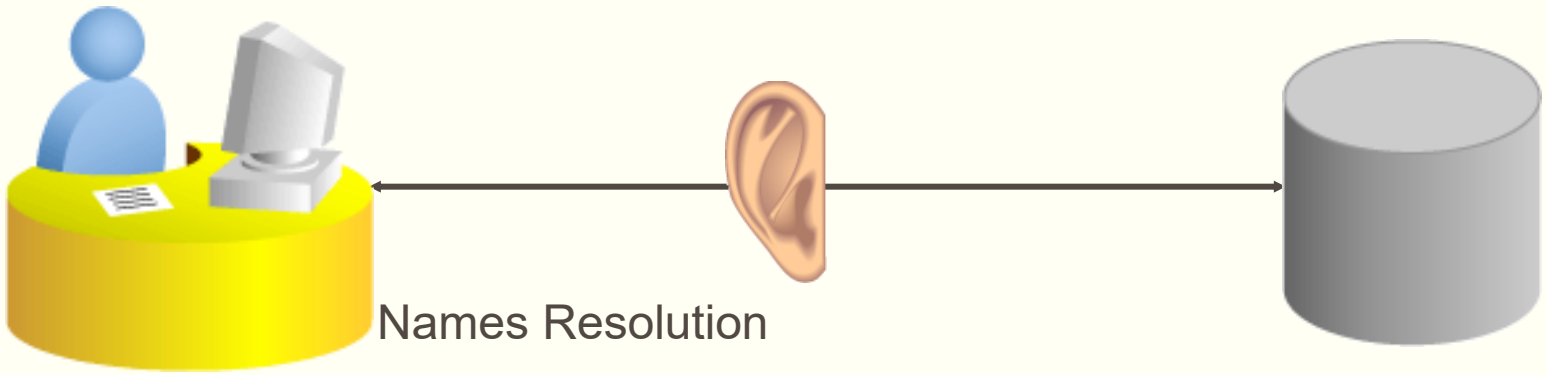
Prompt syntax:

```
LSNRCTL> <command name>
LSNRCTL> start
LSNRCTL> status
```

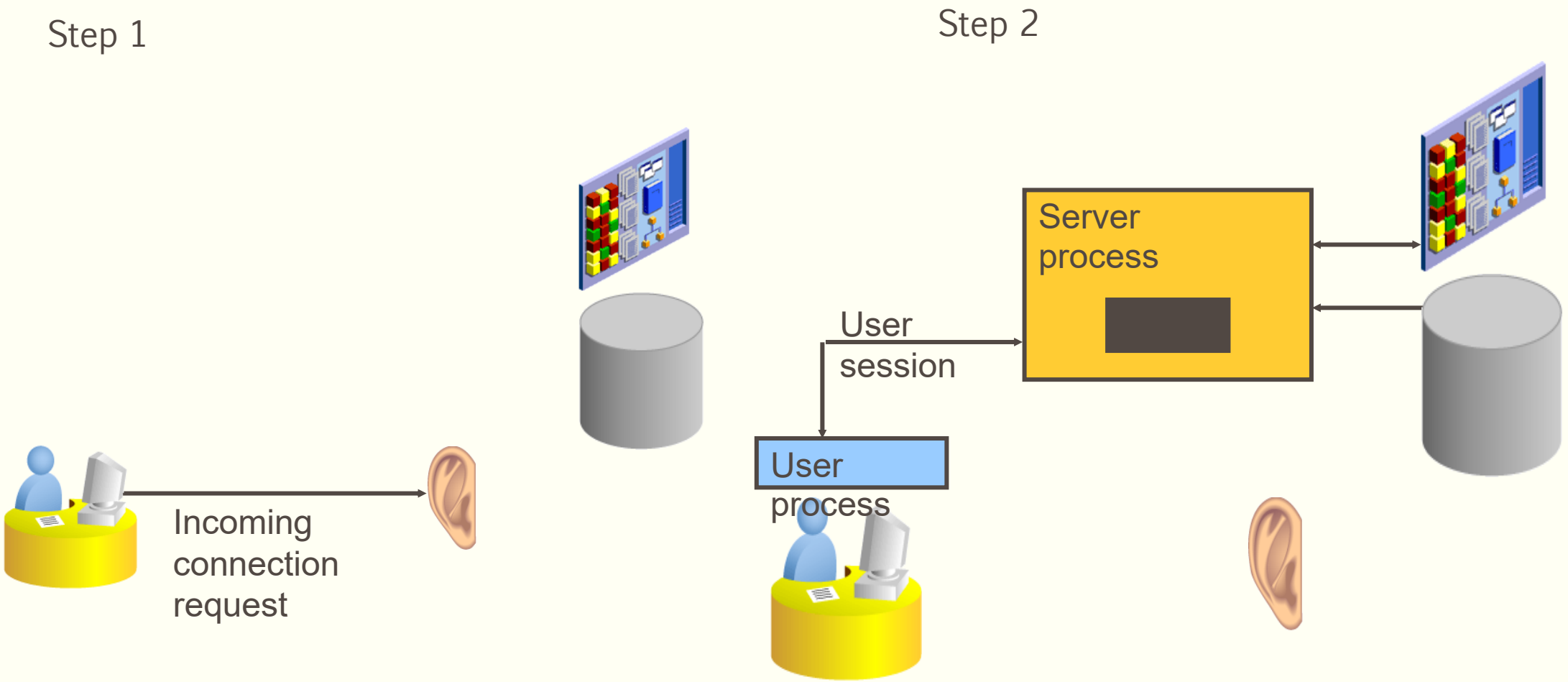
Details Required to Establishing a Database Connection

To make a client or middle-tier connection, Oracle Net requires the client to know the:

- Host where the listener is running
- Port that the listener is monitoring
- Protocol that the listener is using
- Name of the service that the listener is handling



Connection Establishment Process



Connection String

- Users initiate a connection request by providing a connect string.
- A connect string includes a username and password, along with a connect identifier.

Example connection 1

CONNECT scott/tiger@(DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)

(HOST=sales-server1)(PORT=1521))

(CONNECT_DATA=(SERVICE_NAME=sales.us.acme.com)))

Example connection 2

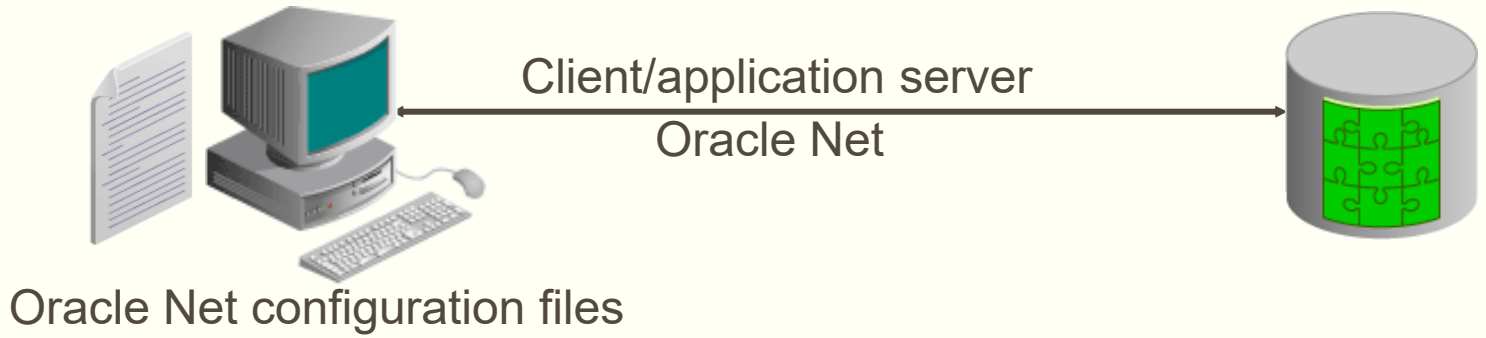
CONNECT scott/tiger@sales

Naming Methods

A **naming method** is a resolution method used by a client application to resolve a connect identifier to a connect descriptor when attempting to connect to a database service.

Oracle Net supports several methods of resolving connection information:

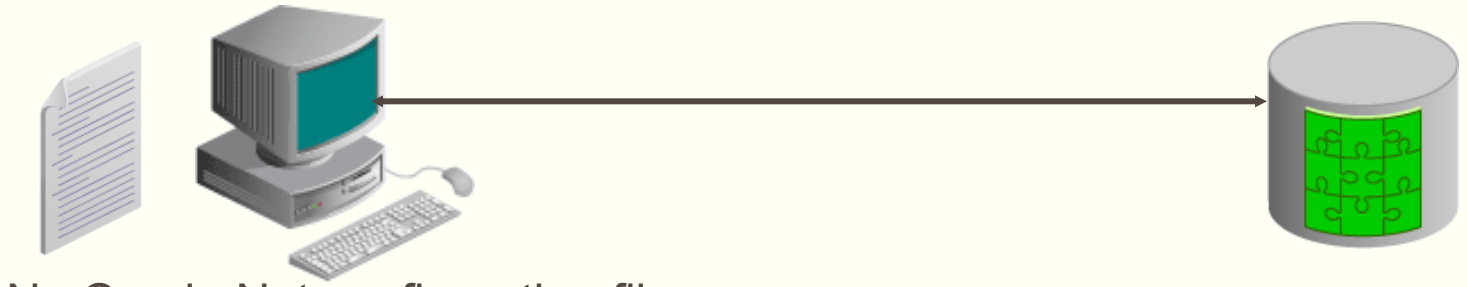
- Easy connect naming: Uses a TCP/IP connect string
- Local naming: Uses a local configuration file
- Directory naming: Uses a centralized LDAP-compliant directory server
- External naming: Uses a supported non-Oracle naming service



Easy Connect

- Is enabled by default
- Requires no client-side configuration
- Supports only TCP/IP (no SSL)
- Offers no support for advanced connection options, such as:
  - Connect-time failover
  - Source routing
  - Load balancing

```
sqlplus hr/hr@db.us.oracle.com:1521/dba10g
```



Local Naming

- Requires a client-side Names Resolution file (tnsnames.ora)
- Supports all Oracle Net protocols
- Supports advanced connection options, such as:
  - Connect-time failover
  - Source routing
  - Load balancing



Sample tnsnames.ora File

```
PDBORCL =
   (DESCRIPTION =
      (ADDRESS_LIST =
         (ADDRESS = (PROTOCOL = TCP)(HOST = ec2-23-21-365-129.compute-1.amazonaws.com)(PORT = 1521))
      )
      (CONNECT_DATA =
         (SERVER = DEDICATED)
         (SERVICE_NAME = pdborcl)
      )
   )
```

Testing Oracle Net Connectivity

The `tnsping` utility that tests Oracle Net service aliases:

- Ensures connectivity between the client and the Oracle Net Listener
- Does not verify that the requested service is available
- Supports Easy Connect Names Resolution:

```
tnsping db.us.oracle.com:1521/dba10g
```

- Supports local and directory naming:

```
tnsping orcl
```

Question

- Mention steps involved to establish a client session

Answer:

1. The client initiates a connect request by providing a connect identifier.
2. The connect identifier is resolved to a connect descriptor by a naming method. This information is returned to the client.
3. The client makes the connection request to the address provided in the connect descriptor.
4. A listener receives the request and directs it to the appropriate database server.
5. After success verification of username and password, the connection is accepted by the database server.

References

- Oracle 19c Admin Guide.